

MATH 3312, SPRING 2026: HOMEWORK 5

- (1) Let R be a commutative ring and let $I \leq R$ be an ideal.

(a) Show that the assignment $J \mapsto J/I$ is a bijection

$$\{\text{Ideals of } R \text{ containing } I\} \leftrightarrow \{\text{Ideals of } R/I\}.$$

(b) Show that, given ideals $I \leq J \leq R$, there is an isomorphism of rings

$$R/J \xrightarrow{\cong} (R/I)/(J/I).$$

Definition 1. A proper ideal $I \leq R$ in a commutative ring is **prime** if for all $a, b \in R$, $ab \in I$ implies that either $a \in I$ or $b \in I$. It is **maximal** if there is no larger proper ideal of R containing I .

- (2) Show that the following are equivalent for a proper ideal $I \leq R$:

(a) I is prime.

(b) R/I is an integral domain.

- (3) Show that the following are equivalent for a proper ideal $I \leq R$:

(a) I is maximal.

(b) R/I is a field.

Hint for both problems: Use Problem 1 to reduce to the case where $I = (0)$ is the zero ideal.

A finitely generated k -vector space V will be called a **finite dimensional k -vector space**, and its **k -dimension** is the unique integer d such that V is isomorphic to k^d . We will denote this integer by $\dim_k V$.

- (4) Suppose that $T : V \rightarrow W$ is a linear transformation of finite dimensional k -vector spaces. Show that

$$\dim_k(\ker T) + \dim_k(\text{im } T) = \dim_k V.$$

*This is usually referred to as **rank-nullity**.*

- (5) Find all polynomials of degree 6 over k that vanish at 0, 1 and 2.

Hint: Let $P_6(k)$ be the space of polynomials of degree at most 6. Then you're computing the kernel of a particular linear transformation from P_6 to k^3 . Note that when k has characteristic 2, you might have a different answer than in general.

Suppose that $k \leq L$ is an extension of fields (that is, one field contained in another as a sub-field). We will say that L is a **finite extension of k** if L is a finite dimensional k -vector space. In this case, we will set $[L : k] = \dim_k(L)$, and call it the **degree** of L over k .

- (6) Suppose that $k \leq L$ is a finite field extension of degree n . Show that every finite dimensional L -vector space V is finite dimensional as a k -vector space, and that

$$\dim_k V = n \dim_L V.$$

- (7) Use the previous problem to prove the following facts, which we saw in a different way earlier this term:

- (a) If k is a finite field of characteristic p , then $|k| = p^r$ for some prime p and $r \in \mathbb{Z}_{\geq 1}$.
 (b) If $k \leq L$ is an extension of finite fields with $|k| = p^r$ and $|L| = q^s$ then $p = q$ and $r \mid s$.

An n -tuple $(v_1, \dots, v_n)$ in a k -vector space V is **k -linearly independent** (or simply **linearly independent**) if the only solution in $a_1, \dots, a_n \in k$ for the equation $a_1 v_1 + \dots + a_n v_n = 0$ is $a_1 = a_2 = \dots = a_n = 0$. The n -tuple is a **basis** if $\{v_1, \dots, v_n\}$ generates V as a k -vector space.

- (8) Let $T : k^n \rightarrow V$ be the unique linear transformation defined by $T(e_i) = v_i$. Show:
 (a) T is injective if and only if $(v_1, \dots, v_n)$ is linearly independent.
 (b) T is an isomorphism if and only if $(v_1, \dots, v_n)$ is a basis.
 (c) If $\dim_k V = n$, then the following are all equivalent:
 (i) $(v_1, \dots, v_n)$ is a basis.
 (ii) $(v_1, \dots, v_n)$ is linearly independent.
 (iii) $\langle \{v_1, \dots, v_n\} \rangle_k = V$.
- (9) Suppose that $k \leq L$ is a field extension and that W is an L -vector space. Is it true that every k -linearly independent subset of W is also L -linearly independent? If true, provide a proof; if false, provide a counter-example.

Suppose that V is a finite dimensional k -vector space and $\mathcal{B} = (v_1, \dots, v_d)$ is a basis for V . Let $\psi_{\mathcal{B}} : k^d \xrightarrow{\sim} V$ be the isomorphism associated with this basis. The **$\mathcal{B}$ -matrix** for a linear transformation $T : V \rightarrow V$, denoted $[T]_{\mathcal{B}}$ is the $d \times d$ -matrix that represents the composition

$$k^d \xrightarrow[\simeq]{\psi_{\mathcal{B}}} V \xrightarrow{T} V \xrightarrow[\simeq]{\psi_{\mathcal{B}}^{-1}} k^d$$

- (10) Let $f(x) \in k[x]$ be a non-constant monic polynomial of degree d . Let $\mathcal{B}$ be the basis for $k[x]/(f(x))$ given by the cosets $(1+(f(x)), x+(f(x)), \dots, x^{d-1}+(f(x)))$. Compute the $\mathcal{B}$ -matrix for the multiplication-by- x linear transformation

$$T : k[x]/(f(x)) \xrightarrow{r(x)+(f(x)) \mapsto xr(x)+(f(x))} k[x]/(f(x)).$$